1. DUAL-MCU (ESP32 + ATTINY1604) & LEVEL SHIFTER

U2
esp32-wemos-d1-mini
IO_00 VCC_(USB) → +5V_LOCAL
IO_02 3V3 → +3V3
IO_04 GND → GND
IO_05/D8 GND
IO_09/SD2 GND
IO_10/SD3 RST
IO_12/TDI NC
IO_13/TCK NC
IO_14/TMS
IO_16/D4
IO_17/D3 CMD
IO_18/D5 TXD
IO_19/D6 RXD
IO_32 CLK
IO_33 SDO
IO_34 SD1
IO_35 TDO
IO_36/SVP/A0
IO_39/SVN
IO_21/D2/SDA
IO_22/D1/SCL
IO_23/D7
IO_25
IO_26/D0
IO_27
ESP32
ESP_RX
ESP_TX

U3
LOGIC_LEVEL_2CH_3V3_5V
+3V3 — LV_3V3 HV_5V — +5V_LOCAL
ESP_TX — TXI TXO — ATTINY_RX
ESP_RX — RXO RXI — ATTINY_TX
GND — GND_L GND_H — GND

J_UPDI1
UPDI_PROG
+5V_LOCAL — +5V
UPDI — UPDI
GND — GND

U1
ATtiny1604-SS
VCC ← +5V_LOCAL
RESET/PA0 ← UPDI
PB0
PB1
PB2 ← ATTINY_TX
PB3 ← ATTINY_RX
PA1 ← 470R ← LED_DIN
PA2 → R_ND0 → N_D0
PA3 → R_ND1 → N_D1
PA4 → R_ED0 → E_D0
PA5 → R_ED1 → E_D1
PA6 → R_SD0 → S_D0
PA7 → R_SD1 → S_D1
330R
GND

C_ATTINY1 100nF
C_LV1 100nF
C_HV1 100nF

2. 12V USB-C PD INPUT, PROTECTION & 5V 3A STEP-DOWN

#PWR01
PWR_FLAG
J_PWR1
USB_C_PD_12V
+12V_IN
GND
SW_MAIN
SW_SPST
F_MAIN1
3A_12V
Q_SW1
AO4407A
S S S G
D D D D
R_GATE100k
R_GATE47k
D_TVS1
SMAJ12A
C_BULK1 100uF 25V
C_SS1 100nF
+12V_BUS

U_BUCK1
MP2315_MINI_5V_3A
+12V_BUS — IN+ — OUT+ — +5V_LOCAL
GND — IN- — OUT-
C_5V1 100uF
C_5V2 100nF

3. 6-PIN MAGNETIC EDGE CONNECTORS: NORTH, EAST, SOUTH

J_NORTH1
+12V_BUS 5V_1
GND GND_1
N_D0 D0
N_D1 D1
GND GND_2
+12V_BUS 5V_2
MAG_NORTH_6PIN_FEMALE

J_EAST1
+12V_BUS 5V_1
GND GND_1
E_D0 D0
E_D1 D1
GND GND_2
+12V_BUS 5V_2
MAG_EAST_6PIN_FEMALE

J_SOUTH1
+12V_BUS 5V_1
GND GND_1
S_D0 D1
S_D1 D0
GND GND_2
+12V_BUS 5V_2
MAG_SOUTH_6PIN_MALE

4. SMART LED MATRIX 4x4 (64 LEDs): WS2812 2x2 DAISY-CHAIN

GND

MOD1 — +5V_LOCAL, LED_DIN, GND
MOD2
MOD3
MOD4
MOD5 — +5V_LOCAL, LED_R1_TO_R2, GND
MOD6
MOD7
MOD8
MOD9 — +5V_LOCAL, LED_R2_TO_R3, GND
MOD10
MOD11
MOD12
MOD13 — +5V_LOCAL, LED_R3_TO_R4, GND
MOD14
MOD15
MOD16

GND_IN1 GND_OUT1
VCC_IN VCC_OUT
DIN DOUT
GND_IN2 GND_OUT2
WS2812 2x2

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